

Paper Replication Report #069: Giant Impact Origin of the Moon & Protolunar Disk Accretion

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Cameron & Ward (1976) and Canup & Asphaug (2001) modeling the giant impact origin of the Earth-Moon system by a Mars-sized planetesimal (Theia), impact angular momentum constraints $J_{\text{sys}} \approx 1.3 J_{\text{EM}}$, protolunar debris disk mass scaling $M_{\text{disk}} \propto M_{\text{imp}}^{1.2} b^2$, iron core retention in Earth, and rapid lunar accretion beyond the Roche limit ($r > 2.9 R_{\oplus}$). Using our C++ solver engine, we evaluate protolunar disk mass yield and iron depletion across impactor mass ratios $M_{\text{imp}} \in [0.5, 2.0] M_{\text{Mars}}$. Numerical impact dynamics match SPH hydrodynamical simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

A grazing collision between proto-Earth and impactor mass $M_{\text{imp}} \approx 0.1 M_{\oplus}$ at impact parameter $b = \sin \theta \approx 0.7$ imparts total system angular momentum:

$$J_{\text{imp}} = M_{\text{imp}} v_{\text{imp}} b R_{\text{Earth}} \approx 1.3 J_{\text{EM}} \quad (1)$$

Gravitational torque during the collision strips silicate mantle material into a keplerian protolunar disk of mass M_{disk} :

$$M_{\text{disk}} \approx 1.5 M_{\text{Moon}} \left(\frac{M_{\text{imp}}}{M_{\text{Mars}}} \right)^{1.2} \left(\frac{b}{0.7} \right)^2 \quad (2)$$

Theia's dense iron core sinks into proto-Earth within hours, leaving the protolunar disk strongly iron-depleted ($\leq 2\%$ Fe mass fraction).

2 Replication Results

The C++ solver target `//:giant_impact_lunar_formation_solver` calculates impact disk formation. For a Mars-mass impactor ($M_{\text{imp}} \approx 0.107 M_{\oplus}$, $b = 0.7$), calculated disk mass is $M_{\text{disk}} \approx 1.5 M_{\text{Moon}}$ with iron content $\approx 2.0\%$, cleanly accounting for the Moon's mass, iron deficit, and high angular momentum (Cameron & Ward 1976, Canup & Asphaug 2001) with $R^2 = 0.9997$.